

Lecture 8 Statistics & its types

Statistics is a Science of collecting, organising and analyzing data

Data = "facts and pieces of information"

Eg Heights of students in class in cm
Gender of person visiting a doctor

Statistics

Descriptive Stats.

Inferential Stats

Definition: It consists of organising & summarizing of data

It consists of using data you have measured to form conclusion.

Sampling to conclusion.

① Measure of central tendency
(mean, mode, median)

① Z test

② t test

③ Chi square

④ ANOVA

② Measure of Dispersion
(Variance & std deviation)

③ Histogram, Pdf, P mass fun
Probability distribution function

Eg Let's consider there are 20 classrooms
and let's say we have calculated ages of
students in one class

→ Ages class: {19, 21, 18, 34, 22, 21, 28, 24}

Descriptive Data

Common age
(mean)

Inferential Data

Are the ages similar
to what you expect
of ages of student
in university

Sample → university

Compare